



GROUP 28 · CSIT321

# Sprint Progress Against the A2 Plan

Certacito.ai · A4 Prototype Presentation appendix

**Project** Certacito.ai, AI agent governance platform

**Document** sprint-progress.md

**Date** 07 August 2026

**Live system** <http://20.92.93.30>

A4 appendix. Status as of 07 August 2026.

A2 committed the group to Agile Scrum with MoSCoW prioritisation across six sprints. This document reports what each of those sprints produced, against the deliverables A2 named, with a pointer to where each claim can be verified in the running system or the repository.

Deliverable wording in the tables below is quoted from A2 section 5.4 so the comparison is like for like. Status is one of **Done**, **Partial**, **Descoped**, **Blocked** or **Not done**.

### The MVP the sprints were building toward

Quoted from A2 section 7.2:

*A governance harness for agentic AI systems that intercepts agent tool calls before execution, enforces a deterministic policy engine, and produces a tamper-evident audit trail suitable for regulatory inspection on Microsoft Azure with role-based access control, human-in-the-loop approval workflows, and a real-time governance dashboard.*

Every clause of that definition is implemented and demonstrable. The sections below show which sprint each clause came from and where it can be verified.

### Sprint 1 (weeks 1-2) - Requirements, architecture, environment

| A2 DELIVERABLE                           | STATUS   | EVIDENCE                                                                                                                                                                                                                                     |
|------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Architecture decision records documented | Not done | No ADR set exists. The decisions themselves were made and are written up in prose in docs/architecture.md (fail-closed at four points, hash preimage covering every column, VM over App Service), but not in the ADR format A2 committed to. |
| Azure environment set up                 | Done     | Ubuntu 24.04 VM in australiaeast, Docker Compose running Postgres 16, Redis 7 and the application image. Runbook in docs/azure-deployment.md.                                                                                                |
| GitHub Projects board running            | Not done | No board is attached to the repository. Task tracking happened in the group chat and in the sprint structure itself.                                                                                                                         |
| Group 2 API contract agreed              | Blocked  | Never received. Tracked as FR-11 in the RTM.                                                                                                                                                                                                 |

Two of the four Sprint 1 deliverables were not produced. Both are process artefacts rather than system capability, which is why the engineering sprints that follow were not held up by them, but they are absent and are recorded as absent.

## Sprint 2 (weeks 3-4) - Interception layer and RBAC

| A2 DELIVERABLE                                        | STATUS | EVIDENCE                                                                                                                                                                                                               |
|-------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Working interception layer capturing agent tool calls | Done   | POST <code>/api/v1/intercept</code> ( <code>backend/api/interception.py</code> ). Every call is evaluated and logged before the action runs. FR-01.                                                                    |
| RBAC implemented                                      | Done   | Administrator, Analyst, Viewer. JWT plus bcrypt, dependency on every endpoint, agent callers use <code>X-API-Key</code> . Regression tests in <code>tests/backend/test_api.py</code> . FR-04.                          |
| CI/CD pipeline running                                | Done   | <code>.github/workflows/ci.yml</code> runs the full suite plus a frontend typecheck and build on every push; <code>deploy.yml</code> deploys to the VM over SSH with a forced-command key. Both verified green. FR-07. |

## Sprint 3 (weeks 5-6) - Policy engine and semantic guardrails

| A2 DELIVERABLE                                        | STATUS                | EVIDENCE                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Policy engine enforcing at least two governance rules | Done, exceeds         | 12 rules in <code>backend/policies.yml</code> , plus a 10 rule pre-built library in <code>rule_library.yml</code> . Conditions are parsed, not keyword matched. FR-02, FR-10.                                                                                                   |
| Semantic guardrail detecting off-topic prompts        | Done                  | <code>backend/services/semantic_guard.py</code> , wired ahead of the policy engine and exposed at <code>/api/v1/guardrails/check</code> . Detects instruction override and exfiltration patterns; the live demo scenario scores its injection attempt at 94% confidence. FR-08. |
| Human approval queue working                          | Done, with limitation | High and Critical escalations pause into <code>approval_queue</code> with an SLA deadline; reviewer identity comes from the JWT. <b>Limitation:</b> SLA expiry is evaluated when the queue is read rather than by a background worker. FR-05.                                   |

## Sprint 4 (weeks 7-8) - Audit store, dashboard, Databricks

| A2 DELIVERABLE                                        | STATUS | EVIDENCE                                                                                                                        |
|-------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------|
| End-to-end audit trail with tamper-evident guarantees | Done   | SHA-256 chain over every stored column, walked from a fixed genesis by GET <code>/api/v1/audit/verify</code> , which recomputes |

|                                           |                 |                                                                                                                                                                                                                                                       |
|-------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           |                 | each hash and names the first bad entry. Tamper tests in <code>tests/backend/test_audit.py</code> . FR-03.                                                                                                                                            |
| Governance dashboard with real-time feeds | <b>Done</b>     | React dashboard on a JWT-authenticated WebSocket with a polling fallback. Live feed, KPI tiles, risk breakdown, top violations. FR-06.                                                                                                                |
| Databricks migration if access confirmed  | <b>Descoped</b> | A2 made this conditional on access being confirmed. Access was never confirmed, so the audit store stays on PostgreSQL. The hash chain provides the tamper-evidence the migration was wanted for, which is why this was descoped rather than carried. |

## Sprint 5 (week 9) - Integration testing and red team

| A2 DELIVERABLE                                 | STATUS          | EVIDENCE                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group 2 integration scenario tested end to end | <b>Blocked</b>  | Depends on an API contract from another group that was never provided. <code>POST /api/v1/demo/healthcare-scenario</code> exercises the same end-to-end path against simulated GP office actions, with all five steps going through the real engine and the real semantic guard. It is a stand-in, not the integration. FR-11. |
| Kali Linux red team testing complete           | <b>Not done</b> | No red team exercise was run. The adjacent evidence is the semantic guard test suite, which covers injection, jailbreak and exfiltration patterns, and the RBAC regression tests. Neither is a substitute for adversarial testing by a person.                                                                                 |
| DLP and risk classification if capacity allows | <b>Partial</b>  | Risk classification is done ( <code>backend/services/risk_classifier.py</code> , FR-09). Data loss prevention beyond top-level PII masking was not built. A2 marked both as capacity-permitting.                                                                                                                               |

## Sprint 6 (week 10) - Documentation and demonstration

| A2 DELIVERABLE                      | STATUS          | EVIDENCE                                                                                                                                                                                                                                   |
|-------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complete technical documentation    | <b>Done</b>     | Seven appendix documents in <code>docs/</code> : architecture (with system and entity relationship diagrams), functionality, interface design, branding and style guide, requirements traceability, contribution table, and this document. |
| User guide for compliance operators | <b>Not done</b> | No operator-facing guide exists. <code>docs/live-demo-cheatsheet.md</code> covers the same screens from a defending-the-system angle rather than an operator angle.                                                                        |
| A3/A4 deliverables                  | <b>Done</b>     | This appendix set, the prototype presentation, and the working system.                                                                                                                                                                     |
| Oral defence preparation            | <b>Done</b>     | Speaking script and a question and answer cheatsheet covering the interception path, fail-closed behaviour, the hash chain, the semantic guard, the policy engine, RBAC, API keys, approvals and reporting.                                |

## Summary

| SPRINT | DELIVERED | NOT DELIVERED                                      |
|--------|-----------|----------------------------------------------------|
| 1      | 1 of 4    | ADRs, Projects board, Group 2 contract (external)  |
| 2      | 3 of 3    | none                                               |
| 3      | 3 of 3    | none                                               |
| 4      | 2 of 3    | Databricks (descoped on A2's own condition)        |
| 5      | 0.5 of 3  | Group 2 integration (external), Kali red team, DLP |
| 6      | 3 of 4    | operator user guide                                |

Every deliverable that was a piece of the system itself was built. The gaps sit in two places: process artefacts the team skipped in favour of building (Sprint 1), and work that depended on something outside the team's control (Group 2) or was explicitly conditional in A2 (Databricks, DLP). The red team exercise is the exception: it was in scope, unconditional, and not done.

### Reflection: was there enough time to learn the stack?

A2 asked the group to reflect on this. The honest answer is that the stack was learnable but the cost showed up in specific, traceable places rather than as a general slowdown, and all three of these were found late:

- **bcrypt and passlib.** Every login on a clean install broke because passlib probes the backend with an over-length string that bcrypt 4.1 and later raise on rather than truncating. The fix was to drop passlib and call bcrypt directly. Time lost to a library interaction nobody had met before.
- **Vite does not typecheck.** The frontend build script only bundles. A component that referenced a variable which was never passed into it built cleanly and passed CI, and crashed the console the moment anyone signed in. CI now runs `tsc --noEmit` before the build. This is exactly the class of mistake a team that had used the toolchain before would not have shipped.
- **A missing lockfile.** The production image runs `npm ci`, which refuses to install without `package-lock.json`. CI used `npm install`, which does not

need one, so CI stayed green over a tree whose production image could not build at all. CI now runs `npm ci` so it fails the same way the image does.

The common thread is that the team's checks agreed with each other while disagreeing with reality. Given the same ten weeks again, the useful change would not be more time on the language or the framework. It would be making the pipeline exercise the same paths as production from Sprint 2, so a green badge means something.